



```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: # Create a Dataset With Wrong format, Missing value And Duplicate value
```

```
In [9]: data = {
    'Employee_ID': [101, 102, 102, 103, np.nan, 105],
    'Name': ['Aman', 'Rahul', 'Rahul', 'Sanya', 'Vikram', 'Priya'],
    'Join_Date': ['2026-01-15', '2026-02-20', '2026/02/20', '15-03-2026', '202',
    'Salary': [50000, 60000, 60000, np.nan, 75000, 80000],
    'Department': ['IT', 'HR', 'HR', 'Sales', 'IT', 'HR']
}
```

```
In [4]: # 1. Create a dataframe And save In Csv File
```

```
In [10]: df = pd.DataFrame(data)
df.to_csv('raw_dataset.csv', index=False)
print("--- Raw Dataset ---")
print(df)
```

--- Raw Dataset ---

	Employee_ID	Name	Join_Date	Salary	Department
0	101.0	Aman	2026-01-15	50000.0	IT
1	102.0	Rahul	2026-02-20	60000.0	HR
2	102.0	Rahul	2026/02/20	60000.0	HR
3	103.0	Sanya	15-03-2026	NaN	Sales
4	NaN	Vikram	2026-04-12	75000.0	IT
5	105.0	Priya	2026-05-18	80000.0	HR

```
In [11]: # 2. Data Cleaning And Preprocessing Pipeline Function
```

```
In [12]: def clean_data_pipeline(file_path):
    df = pd.read_csv(file_path)
    return df
path = r"C:\Users\Admin\Desktop\raw_dataset.csv"
my_df = clean_data_pipeline(path)
print(my_df.head())
```

	Employee_ID	Name	Join_Date	Salary	Department
0	101.0	Aman	2026-01-15	50000.0	IT
1	102.0	Rahul	2026-02-20	60000.0	HR
2	102.0	Rahul	2026/02/20	60000.0	HR
3	103.0	Sanya	15-03-2026	NaN	Sales
4	NaN	Vikram	2026-04-12	75000.0	IT

```
In [13]: # Remove Duplicate Value
df.drop_duplicates(inplace=True)
df
```

Out[13]:

	Employee_ID	Name	Join_Date	Salary	Department
0	101.0	Aman	2026-01-15	50000.0	IT
1	102.0	Rahul	2026-02-20	60000.0	HR
2	102.0	Rahul	2026/02/20	60000.0	HR
3	103.0	Sanya	15-03-2026	NaN	Sales
4	NaN	Vikram	2026-04-12	75000.0	IT
5	105.0	Priya	2026-05-18	80000.0	HR

```
In [14]: # Handle Missing value in salary column.
if df['Salary'].isnull().sum() > 0:
    salary_median = df['Salary'].median()
    df['Salary'] = df['Salary'].fillna(salary_median)
# Handle Missing value in Employee_ID
df.dropna(subset=['Employee_ID'], inplace=True)
df['Employee_ID'] = df['Employee_ID'].astype(int)
df
```

Out[14]:

	Employee_ID	Name	Join_Date	Salary	Department
0	101	Aman	2026-01-15	50000.0	IT
1	102	Rahul	2026-02-20	60000.0	HR
2	102	Rahul	2026/02/20	60000.0	HR
3	103	Sanya	15-03-2026	60000.0	Sales
5	105	Priya	2026-05-18	80000.0	HR

```
In [17]: # Correct Extra Space.
df['Department'] = df['Department'].str.strip()
df
```

Out[17]:

	Employee_ID	Name	Join_Date	Salary	Department
0	101	Aman	2026-01-15	50000.0	IT
1	102	Rahul	2026-02-20	60000.0	HR
2	102	Rahul	2026/02/20	60000.0	HR
3	103	Sanya	15-03-2026	60000.0	Sales
5	105	Priya	2026-05-18	80000.0	HR

```
In [19]: def clean_date_pipeline(df):
df['Join_Date'] = pd.to_datetime(df['Join_Date'], format='%Y-%m-%d', error
df.dropna(subset=['Join_Date'], inplace=True)
return df
df
```

Out[19]:

	Employee_ID	Name	Join_Date	Salary	Department
0	101	Aman	2026-01-15	50000.0	IT
1	102	Rahul	2026-02-20	60000.0	HR
2	102	Rahul	2026/02/20	60000.0	HR
3	103	Sanya	15-03-2026	60000.0	Sales
5	105	Priya	2026-05-18	80000.0	HR

```
In [26]: # save a new csv file.  
df.to_csv('my_clean_data.csv', index=False)
```

In []:

In []: